Abinav GT

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Skills

Languages: Python, C++, Java, JavaScript
Framework: TensorFlow, scikit-learn, NLTK, FastAPI, Flask, HTML, CSS
Tools/Platforms: OpenCV, MediaPipe, Power BI, Excel, MYSQL, Git, GitHub Actions
Soft Skills: Problem-Solving, Team Player, Time Management, Adaptability

Training

Code Quest: DSA Summer Bootcamp — Jun' 25 – Jul' 25

- Built strong foundations in **Data Structures** and solved **advanced algorithmic** problems covering Recursion, Dynamic Programming, Greedy Algorithms, and Graph Traversals
- Improved problem-solving efficiency by practicing coding problems on platforms like **Leetcode**, focusing on **optimization** and **time** complexity analysis
- Solved **150+** DSA problems, improving speed and accuracy in coding interviews

Projects

Real-Time Facial Emotion Recognition System — Mar' 26

- Built a real-time facial emotion recognition system using a **CNN** model, achieving **~85%** accuracy through data **preprocessing**, data **augmentation** and **hyper parameter** tuning
- Implemented a computer vision pipeline using **OpenCV** and **MediaPipe** for face detection and landmark extraction in real time
- Developed an interactive frontend with real-time emotion visualization with graphical analysis, tracking emotion counts across frame for enhanced analytical insights
  Tech Stack: Python, TensorFlow, scikit-learn, FASTAPI, OpenCV, MediaPipe, Data visualization lib

Heart Stroke Prediction — Feb' 26

- Architected a comprehensive diagnostic engine that leverages a **KNN** algorithm to identify high-risk cardiovascular patterns, achieving **~86% F1-score** through cross validation
- Engineered a robust data **preprocessing** workflow involving **Standard scaler** for feature **normalization** and **One Hot** Encoding for categorical health metrics
- Built an interactive web interface using **Streamlit** that provides real-time health risk assessments, visualizing user metrics and diagnostic confidence levels
  Tech Stack: Python, TensorFlow, scikit-learn, joblib, Streamlit, Data preprocessing and visualization

Currency Denomination Calculator (ATM Simulator) — Jul' 25

- Built a web based ATM simulator that calculates the minimum number of currency notes required for a given withdrawal amount using a **Greedy Algorithm**
- Created a responsive user interface by integrating a **Flask**-based Python enabling seamless asynchronous data processing, data fetching of a user and real-time money calculations
- Architected a persistent data management system using **MySQL** to track transaction patterns and **deployed** the final solution via an automated CI/CD pipeline to ensure high availability and accessibility.
  Tech Stack: Python, JavaScript, Flask, MySQL, HTML, CSS

Certificates

- Data Science and Analysis | **Vital Skills - IIT Kanpur** — Jan' 25
- Core Java and Basic Application Development | **Internz Learn** — Apr' 24

ACHIEVMENTS

- Engaged in **CNCF** community **meetups** and Participated in **ML** focused **hackathons** — Feb' 26
- Ranked **15th** in Hack Quest – 24 Hours CTF Challenge organized by LPU and **UpGrad** campus — Apr' 24

Education

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology – Computer Science and Engineering; **CGPA: 7.01** — Aug' 23 – Present

Happy Valley School — Vijayawada, Andhra Pradesh
Intermediate (CBSE) - **Percentage: 88%** — Apr'20 – Mar'22